

Qazi Saaheelur Rahaman

PHD STUDENT | COMPUTATIONAL BIOPHYSICS AND COMPLEX SYSTEMS

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Education

Aix-Marseille University / SCHADOC Doctoral School

PHD – DEEP-LEARNING TENSORS OF CELL REARRANGEMENTS (DELETE)

Marseille, France

2025 – Present

Indian Institute of Science Education and Research (IISER) Berhampur

BS-MS DUAL DEGREE, CPI: 7.9

Berhampur, India

Dec. 2020 – Aug. 2025

Research Interests

Computational Biophysics | Cell and Tissue Dynamics | Active Matter | Image Analysis
Complex Systems | Adaptive Networks | Collective Dynamics | Machine Learning

Current Research

DEep-LEARNING TENSors of cell rearrangements (DELETE)

DOCTORAL RESEARCHER

Marseille, France

2025 – Present

- Developing quantitative and computational approaches to characterize cell and tissue rearrangements from imaging data.
- Combining image analysis, physical modelling and machine-learning tools to study collective dynamics and structural transitions in biological systems.

Publications

Collective dynamics of multiplex networks with distinct adaptation rules

PHYSICAL REVIEW E 112, 064313

2025

- Akash Yadav, **Qazi Saaheelur Rahaman**, V. K. Chandrasekar, and D. V. Senthilkumar. DOI: 10.1103/vndm-lcqs.

Heterogeneous nucleation in a multiplex adaptive network

PHYSICAL REVIEW E 111, 054318

2025

- Akash Yadav, **Qazi Saaheelur Rahaman**, V. K. Chandrasekar, and D. V. Senthilkumar. Equal contribution by the first two authors. DOI: 10.1103/PhysRevE.111.054318.

Research Experience

Centre de Physique Théorique, Aix-Marseille University

RESEARCH INTERN

Marseille, France

May 2025 – July 2025

- Worked with the Out-of-Equilibrium Mechanics team on quantitative image analysis of cells and tissues.
- Supervised by Dr. Jean-François Rupprecht, CNRS Researcher.

Indian Institute of Science Education and Research (IISER) Thiruvananthapuram

MASTER'S THESIS STUDENT

Thiruvananthapuram, India

June 2024 – May 2025

- Studied synchronization transitions and collective dynamics in multiplex adaptive networks, including heterogeneous nucleation and distinct adaptation rules.
- Research led to two publications in *Physical Review E*; supervised by Dr. D. V. Senthilkumar.

Institute of Cosmology and Gravitation, University of Portsmouth

RESEARCH INTERN

Remote

Feb. 2023 – Sept. 2023

- Applied machine-learning regression methods to predict dark-matter halo properties from galaxy photometric and simulation-derived features.
- Developed scientific data-analysis workflows in Python under the supervision of Dr. Christopher Lovell.

Selected Scientific Activities

Jun.–Jul. 2026	Poster , ICISE Soft Matter Conference	<i>Vietnam</i>
Jun. 2026	Participant , CENTURI Hackathon	<i>Marseille</i>
Apr. 2026	Participant , SCHADOC Doctoral School Spring School	<i>France</i>
Mar. 2026	Poster , EMBO Conference: AI and Biology	<i>EMBL Heidelberg</i>
Oct. 2025	Workshop , Physical Biology of the Cell workshop by Rob Phillips	

Awards and Scientific Outreach

2022	7th place, Round 1 , International Natural Science Tournament	<i>International</i>
2021	Gold Medal , iGEM 2021; team nominated for Best Diagnostic Project	<i>IISER Berhampur</i>
2021	Invited Student Talk , Turing Patterns – Mat-Sci Student Talk Series, Project Spiral	<i>IISER Berhampur</i>

Skills

Scientific Programming	Python (NumPy, SciPy, pandas, Matplotlib, scikit-learn), C, Java
Computational Methods	Image analysis, machine learning, numerical modelling, network analysis
Research Tools	Jupyter, LaTeX, Git/GitHub, scientific visualization
Design / Media	Adobe Illustrator, Photoshop, Premiere Pro, After Effects

Selected Leadership and Co-Curricular Activities

2021–2024	Coordinator / Design Lead , Episteme – IISER Berhampur Science Magazine	<i>IISER Berhampur</i>
2022–2024	Coordinator / Core Committee , Film Society of IISER Berhampur	<i>IISER Berhampur</i>
2022–2024	Core Committee , 137 Inverse – Physics Club	<i>IISER Berhampur</i>
2021	Gold Medal Team , iGEM 2021 – CRISPR-Cas diagnostic project	<i>IISER Berhampur</i>